Car Accident Injuries: Analysis of Ford's Relationship to Injuries

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Background and Data Cleaning

Motor vehicle safety and crash injury prevention are important concerns for both car manufacturers and society. Some studies suggest that Ford vehicles are associated with a higher number of injury-related crashes, which raises the concerns about whether the brand itself increases those risks. Previous evidence points to potential safety concerns on Ford, including severe mechanical failures on certain models in 2006-2010, large-scale recalls in 2015-2017, and weaker performance in certain crash tests in 2005. Ford also has displayed fire hazards in various models that have the potential to escalate mild accidents to severe ones.

However, an observed relationship does not necessarily imply causation. Other factors, such as driver characteristics, driving behavior, and other environmental and geographical conditions can influence the number of injury-sustained accidents and also car choices. Therefore, on this project, we use a logit regression model to examine whether Ford vehicles are associated with higher probability of injury-related accidents.

In order to prepare the data for analysis, we completed a variety of data cleaning and preparation steps. First, we created the main variables that we will utilize for analysis. This included creating binary columns for necessary data: male, licensed, nystate_driver, unsafe_speed, summer, fall, winter, and driver_inattention, peak_hour, late_night, passengers_in_vehicle, weekend, type_suv_wagon, type_taxi_limo, and type_commercial . This allows the data to be recorded in proper format for a logit regression model. Additionally, the dataset targets 2016 car crashes, so we filtered for only 2016 in the car crash date column. We also handled null values by replacing them with 0 or "Unknown" as seen fit.

Research Question

This analysis aims to answer the following question: Does a car in an automobile accident being a Ford vehicle affect the probability of the occupants sustaining an injury?

Theories & Expected Results

Our hypothesis is that the presence of a Ford vehicle elevates the probability of an injury in a car accident. Ford is the primary regressor, while the remaining variables are included as

controls to reduce potential omitted variable bias. Detailed theoretical motivations and expected relationships are reported in the Appendix Table.

Table - appendix for theories & expected results

Variables	**How it relates to Ford**	**How it relates to probability of crash causing injury**
male	Ford buyers may be more likely male.	Male drivers may engage in riskier driving.
licensed	Licensed drivers are more likely to own vehicles.	Unlicensed drivers may face higher crash risk.
peak_hour	Ford drivers are likely using their vehicle during working hours and commuting to and from work.	More accidents tend to happen during rush hour due to many cars being on the road packed together.
late_night	Ford drivers are more likely to drive late at night than walk/bus/train (10 PM ~ 5 AM).	Driving at night (10 PM ~ 5 AM), when people are tired, under the influence, or sleepy may increase crash probability.
passangers_in_vehicle	Ford may attract users who are family / high-capacity preferred.	More occupants may increase distraction severity.
nystate_driver	Ford drivers are more likely in Southern regions.	Unfamiliar roads may increase risk of injury.
unsafe_speed	Ford vehicles can achieve high speed/performance.	Speedy driving increases injuries risk in a crash.
age_of_vehicle	Based on its long history, Ford vehicles' age have been popular for a long time, increasing the possibility of an older car.	Older cars have fewer safety features.
winter	Ford demand may rise in winter conditions due to its marketing.	Hazardous weather (icy roads, snowstorms) may increase crash risk and injury severity.
driver_inattention	Perceived safety marketing may affect driver inattention.	Distraction increases crash risk and injuries.
summer	Ford travel can peak during the summer season for vacation and road-trip.	Heavy travel volume during vacation and speeding habits can increase the risk of injuries in a crash.
fall	Ford usage can vary due to school return and student travel.	Earlier sunset time and wet roads may increase the chance of a crash and injuries.
weekend	Ford use can differ between working weekdays and leisure weekends.	Weekday congestion may raise crash frequency and injury occurrence.

type_suv_wagon	Ford's SUV and Wagon are popular among their customers.	SUV and Wagon vehicle types can increase injury risk to others due to their large size.
type_taxi_limo	Ford has a historic relationship with the taxi and limo industry with Crown, Lincoln, and Mustangs.	Taxi and Limo drivers can be under increased stress and passengers usually do not wear seatbelts increasing injury risk..
type_commercial	Ford is a dominant force in the commercial vehicle sector with vans, pickup trucks, and heavy trucks	Commercial vehicles are usually much larger and heavier, crashes are typically injury-related for the passenger vehicles; commercial drivers are also often highly fatigued, increasing injury risk.

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===== SUMMARY TABLE =====
             Metric          Value
0        Total Rows  60000.000000
1         Ford Rows   6144.000000
2     Non-Ford Rows  53856.000000
3        Fatal Rows     40.000000
4     Nonfatal Rows  59960.000000
5      Injured Rows  13823.000000
6  Non-Injured Rows  46177.000000
7        Fatal Rate      0.000667
8       Injury Rate      0.230383
9        Ford Share      0.102400

===== SIMPLIFIED VEHICLE TYPE CONTROLS =====
type_suv_wagon     19885
type_taxi_limo      2788
type_commercial     5709
dtype: int64
```

Our dataset's granularity is such that each observation represents a vehicle involved in a crash. The final sample contains 78,555 observations, of which 1,3823 rows contain an injury.

Models

We used the following equation to conduct the model:

ln[p/(1-p)] = B0 + B1*ford* + B2*ageofvehicle* + B3*passangers_in_car* + B4*male* + B5*nysdriver* + B6*UnsafeSpeed* + B7*winter* + B8*driverinattention* + B9*licensed* + B10*peakhour* + B11*late_night* + B12*summer* + B13*fall* + B14*weekend* + B15*type_suv_wagon* + B16*Type_taxi_limo* + B17*type_commercial*

Where p is the probability of an injury in a crash.

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===== MULTIPLE LOGIT REGRESSION: INJURED OUTCOME =====
                           Logit Regression Results
==============================================================================
Dep. Variable:                injured   No. Observations:                60000
Model:                          Logit   Df Residuals:                    59982
Method:                           MLE   Df Model:                           17
Date:                Tue, 05 May 2026   Pseudo R-squ.:                0.02789
Time:                        15:43:06   Log-Likelihood:                -31481.
converged:                       True   LL-Null:                       -32384.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==============================================================================================
                                 coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------------------
const                      -1.4468      0.310     -4.665      0.000      -2.055      -0.839
ford                       -0.0107      0.033     -0.325      0.745      -0.076       0.054
unsafe_speed                0.9678      0.064     15.066      0.000       0.842       1.094
peak_hour                  -0.0009      0.022     -0.042      0.966      -0.044       0.042
late_night_v2               0.2641      0.028      9.349      0.000       0.209       0.320
male                        0.0188      0.023      0.823      0.410      -0.026       0.064
licensed                    0.2238      0.038      5.842      0.000       0.149       0.299
passengers_in_car           0.5542      0.022     24.905      0.000       0.511       0.598
age_of_vehicle              0.0182      0.002     10.315      0.000       0.015       0.022
driver_inattention_v2       0.1195      0.025      4.784      0.000       0.071       0.168
summer                     -0.4286      0.311     -1.378      0.168      -1.038       0.181
winter                     -0.5250      0.309     -1.697      0.090      -1.132       0.082
fall                       -0.4793      0.309     -1.551      0.121      -1.085       0.126
weekend                     0.0061      0.023      0.265      0.791      -0.039       0.051
nystate_driver              0.3099      0.030     10.454      0.000       0.252       0.368
type_suv_wagon             -0.0290      0.022     -1.332      0.183      -0.072       0.014
type_taxi_limo             -0.1345      0.049     -2.723      0.006      -0.231      -0.038
type_commercial            -0.4835      0.040    -12.139      0.000      -0.562      -0.405
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Main Findings

The *ford* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car crash decreases by -0.0107. However, its p-value of 0.745 is greater than 0.05; this means that *ford* is not a statistically significant variable in measuring the probability of injuries in a crash. With this current model we are unable to confidently assert a relationship between a car being a Ford vehicle and an increased probability of an injury being sustained in a car accident. The *unsafe_speed* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car crash increases by 0.9678. Unsafe_speed has a p-value of 0.0, below 0.05, meaning that it is statistically significant. The *late_night_v2* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury increases by 0.2641. The *late_night_v2* variable has a p-value of 0.0, meaning it is statistically significant. The *licensed* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car crash increases by 0.2238. It can be that licensed drivers are more assertive on the road, and may be more likely to be involved in more injury-related crashes, which can lead to injuries. The *licensed* variable has a p-value of 0.0,

meaning it is statistically significant. The *Passengers_in_car* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car increases by 0.5542. *Passengers_in_car* has a p-value of 0.0, meaning it is statistically significant. The *age_of_vehicle* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car increases by 0.0182. The *age_of_vehicle* variable has a p-value of 0.0, meaning it is statistically significant. The *driver_inattention_v2* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car increases by 0.1195. The *driver_inattention_v2* has a p-value of 0.0, meaning it is statistically significant. The *nystate_driver* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car increases by 0.3090. The *nystate_driver* variable has a p-value of 0.0, meaning it is statistically significant. The *type_taxi_limo* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car decreases by -0.1345. The *type_taxi_limo* variable has a p-value of 0.006, meaning it is statistically significant. The *type_commercial* coefficient signifies that when the unit increases by 1 and all other variables are constant, the log odds of an injury in a car decreases by -0.4835. The *type_commercial* has a p-value of 0.0, meaning it is statistically significant.

Limitations

This model cannot be used to imply causality since it has omitted variable bias. For instance, the *Age_Of_Driver* can greatly contribute to the resulting injuries of the passengers. Ford is popular among the younger population for the “American” brand and its versatility for daily commute and for work. Younger drivers are also more likely to have a Ford as they are affordable. Younger drivers are less experienced and are often more careless leading to more injury-related crashes and therefore are more likely to sustain injuries.

Conclusion

Although this model is complete now, the research is nowhere near finished. As stated in the limitations, there are additional core variables that are missing from the model that are leaving the model incomplete as of now. However, according to the current modCel, several variables such as *unsafe_speed*, *late_night_v2*, *licensed*, and *passengers_in_car* have significant correlations with the probability of an injury in a car accident, while the Ford variable itself was not significant in the final model. In the future, we must collect additional data and conduct further research to continue adding core variables to determine causality.